Dispersion Managed Transceiver for Next-Gen AI Fabrics

Cost-Effective 800G (20km) & 1.6T (10km) Interconnects
for Scale-Out, Scale-Across, and OCS SuperPOD Architectures

Grace Zhang, Ph.D.

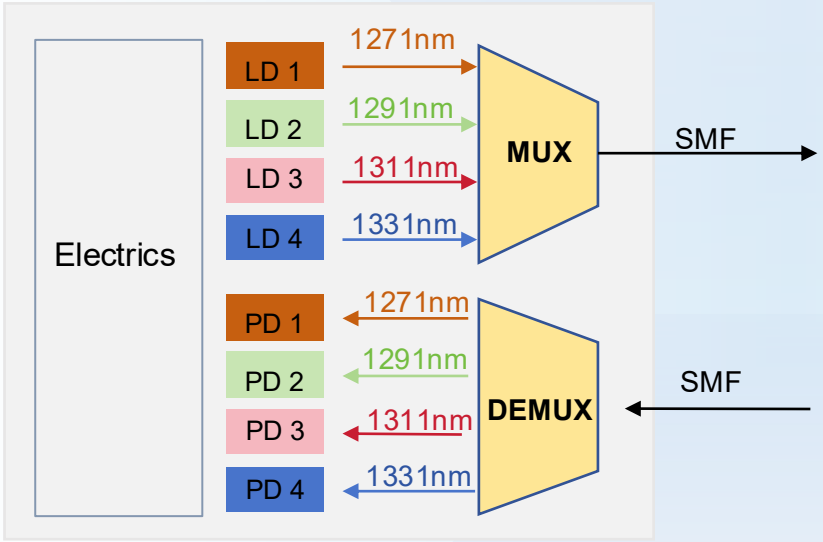
OCP 2026 October 12–15, 2026 San Jose Convention Center San Jose, CA, USA BOOTH #C84

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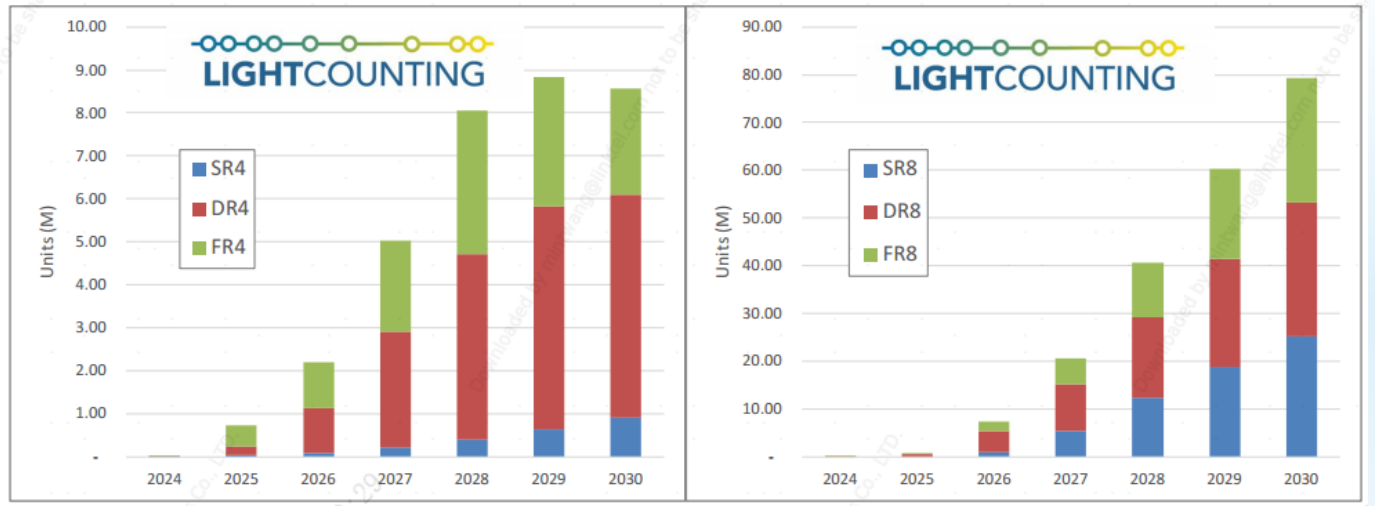
CWDM Remains the Preferred Backbone for High-Density AI Fabrics

- **Why CWDM Dominates**:
  - **Fiber Efficiency**: Multiplexes 4–8 wavelengths over single-mode fiber pairs;
  - **Operational Simplicity**: Fewer fiber end faces mitigate contamination risk and simplify connector handling;
  - **Market Trend:** CWDM FR4/FR8 formats are projected to lead 4x200G and 8x200G volume shipments through 2030.



CWDM4 Transceiver

Figure 5-10: Shipments of 4x200G and 8x200G transceivers by reach

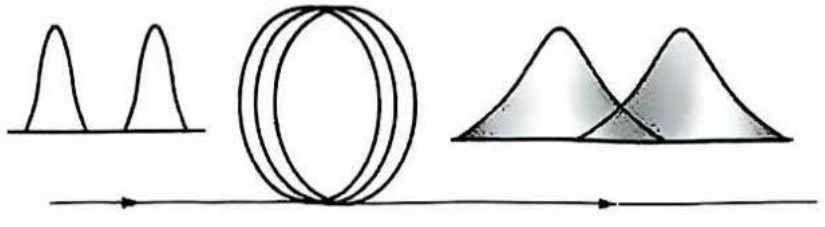


Source: LightCounting

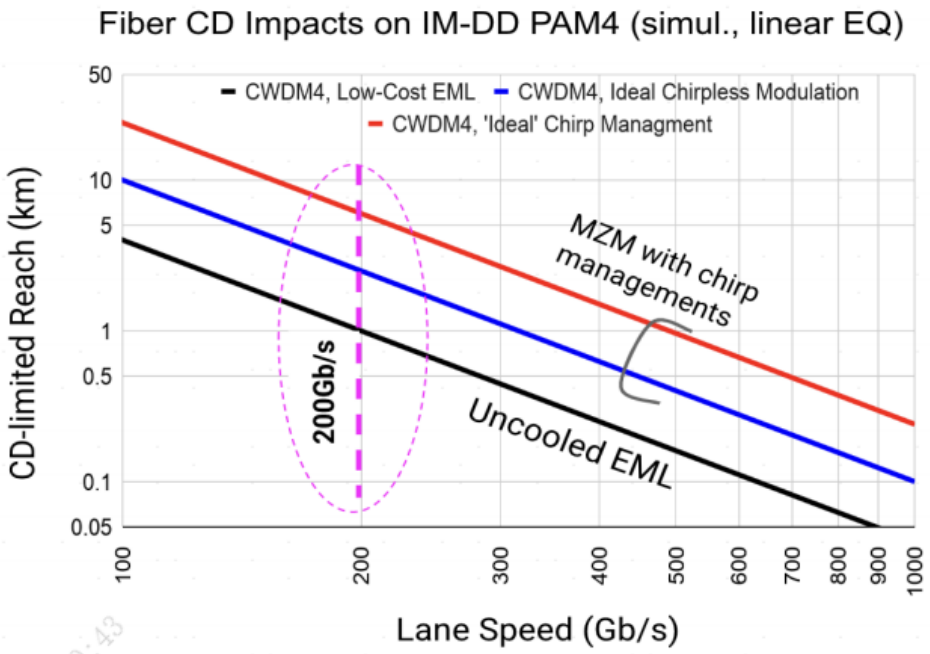
The Fundamental Bottleneck: Fiber Chromatic Dispersion
severely restricts reach as per-lane baud rates increase

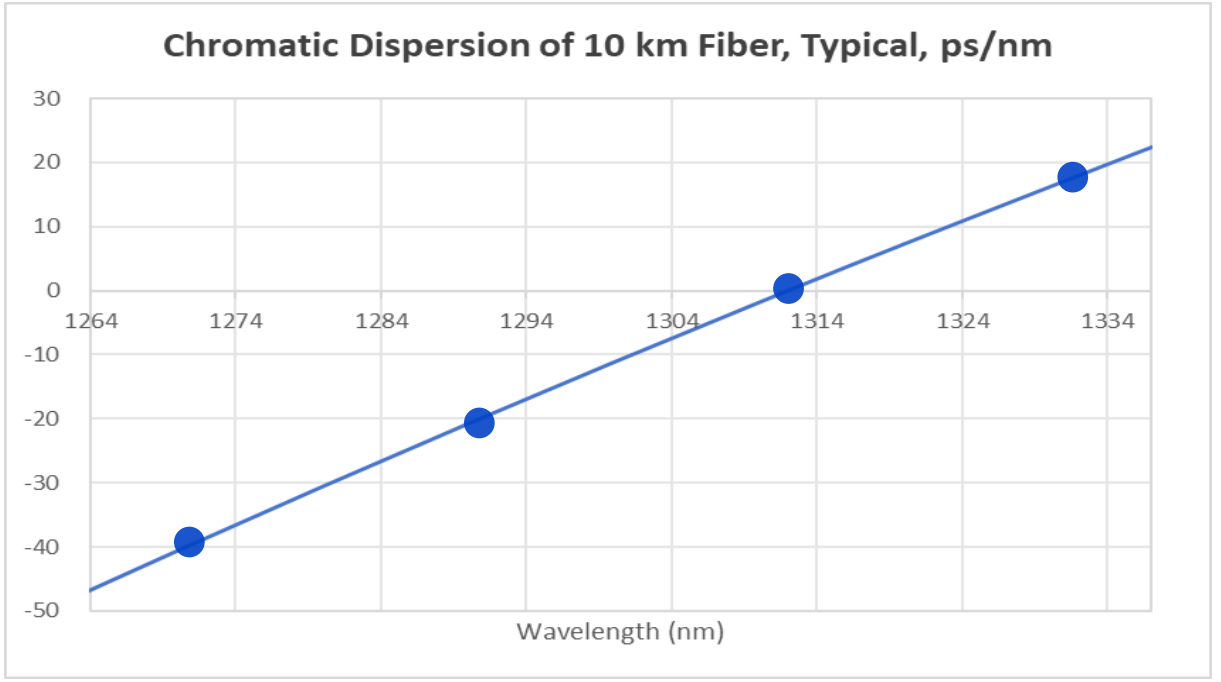
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Dispersion Scaling: 2× Data Rate → ¼ Reach



$$\Delta t_{chrom} = D(\lambda) L \Delta\lambda$$





Reach Breakdown by Lane Speed:

- **100G / lane:** Achieves 10km reach without optical dispersion management.
- **200G / lane:** Unmanaged reach collapses to 2.5km.
- **400G / lane:** Unmanaged reach drops to just 600 meters.

X. Zhou, C. F. Lam, R. Urata and H. Liu, "State-of-the-Art 800G/1.6T Datacom Interconnects and Outlook for 3.2T," 2023 Optical Fiber Communications Conference and Exhibition (OFC), San Diego, CA, USA, 2023, pp. 1-3, doi: 10.1364/OFC.2023.W3D.1.

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Linktel Dispersion-Managed Transceiver Roadmap

- Incorporating Dispersion Managed Device(DMD) in existed CWDM FR4 design
- Cost Effective
- Ignorable Power Consumption

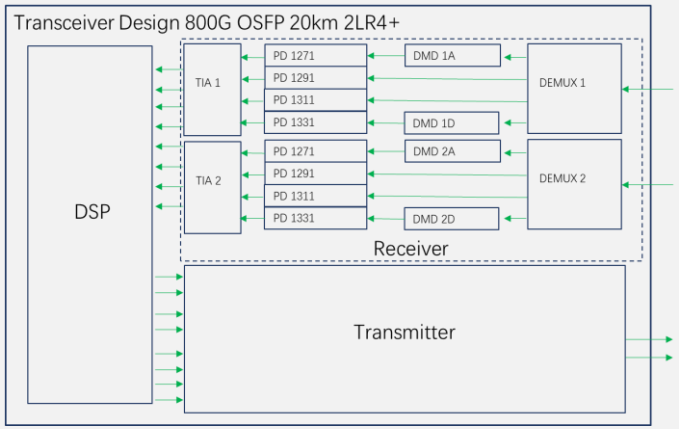


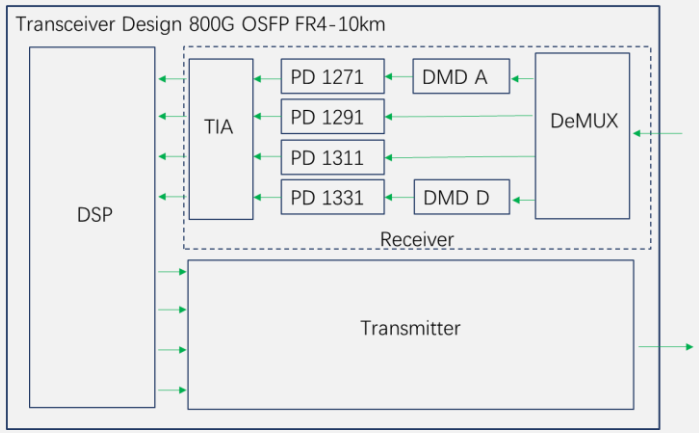
Live Demo: #C84

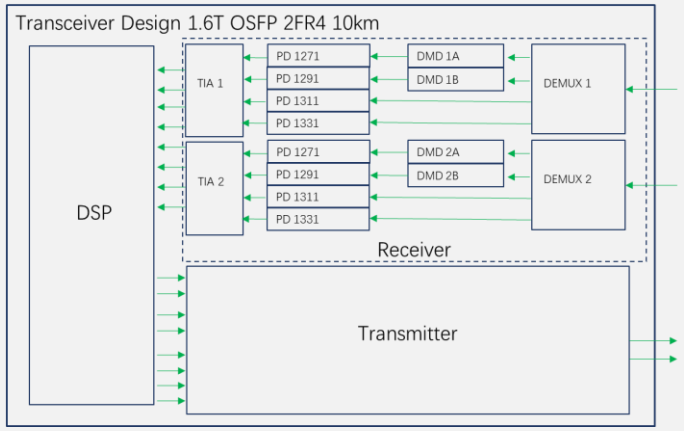
	800G 2LR4+ 20km	800G FR4 10km	1.6T 2FR4 10km
Design	EML(100G/lane CWDM) 4CH DMD	EML(200G/lane CWDM) 2CH DMD	SiPh (200G/lane CWDM) 4CH DMD
Performance	20km BER: e-9, e-10	10km BER: e-10, e-11 (212GB); 10km BER: error free(226GB with inner FEC).	10km BER: e-11, e-12(212GB); 10km BER: error free(226GB with inner FEC).
Schedule	Beta	Beta	Alpha

Lightwave Innovation Award

Lightwave Innovation Award





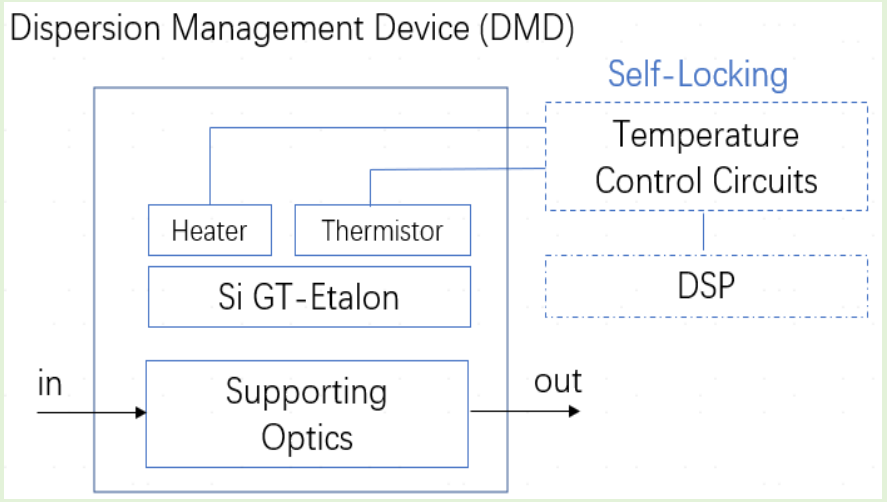


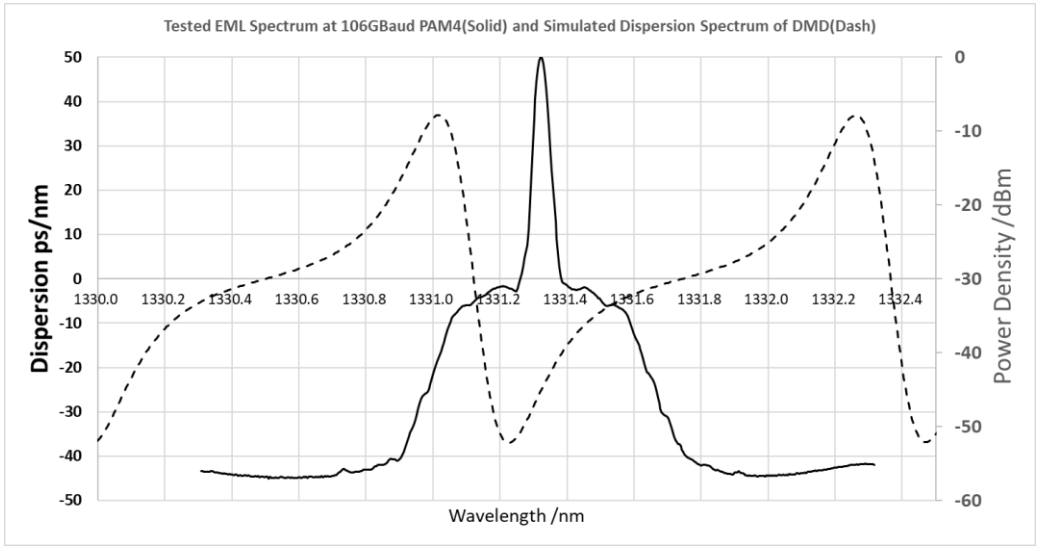
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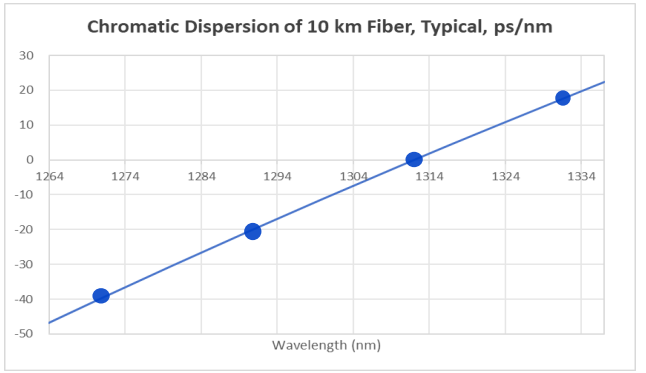
Core Tech Breakdown: Silicon GT-Etalon DMD Architecture

- **Device Architecture:** Single-cavity Silicon Gires-Tournois (GT) Etalon with integrated micro-heater and thermistor.
- **Near-Zero Insertion Loss:** > 99.9%High-Reflectivity (HR) coating eliminates optical loss.
- **Wide Tuning Range:** Provides dynamic dispersion compensation across positive and negative values.
- **DSP Closed-Loop Self-Locking:** Auto-tunes to maximize SNR, dynamically adapting to fiber length variations and transmitter wavelength drift.





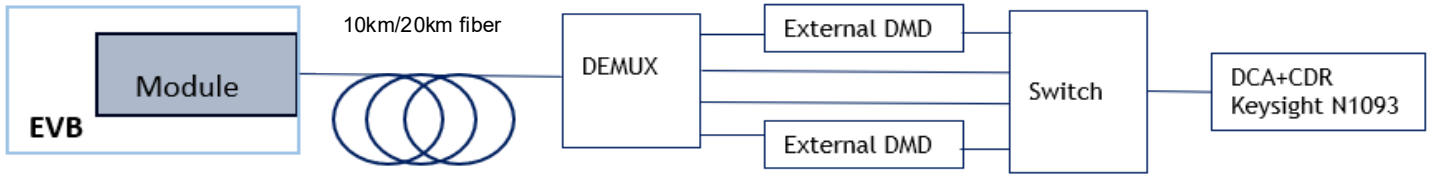




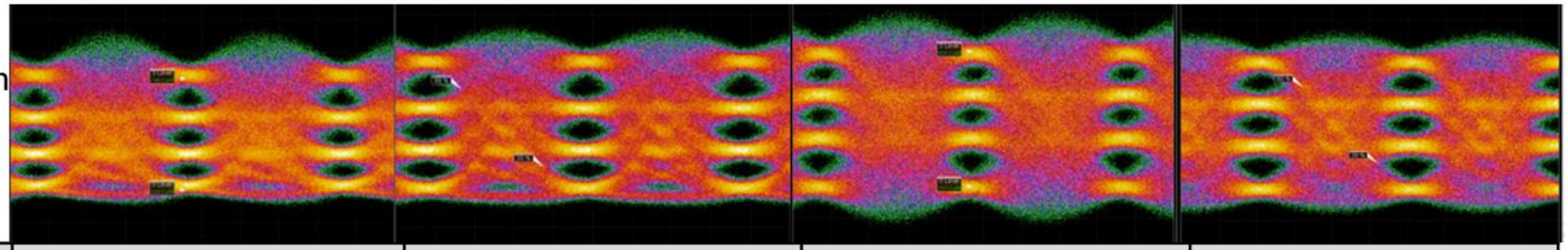
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Transmitter Eye Diagram Recovery

Test setup for TP3 Eye Diagram

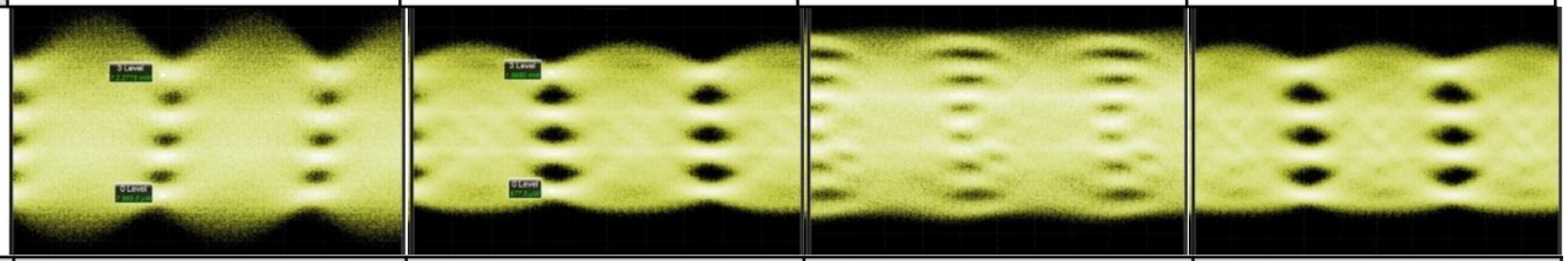


800G 20km 2LR4+



106Gb 20km	1271 without DMD	1271 with DMD	1331 without DMD	1331 with DMD
ER/dB	5.6	5.7	5.2	5.3
TDECQ/dB	NA	1.5	NA	2.5

800G FR4-10km



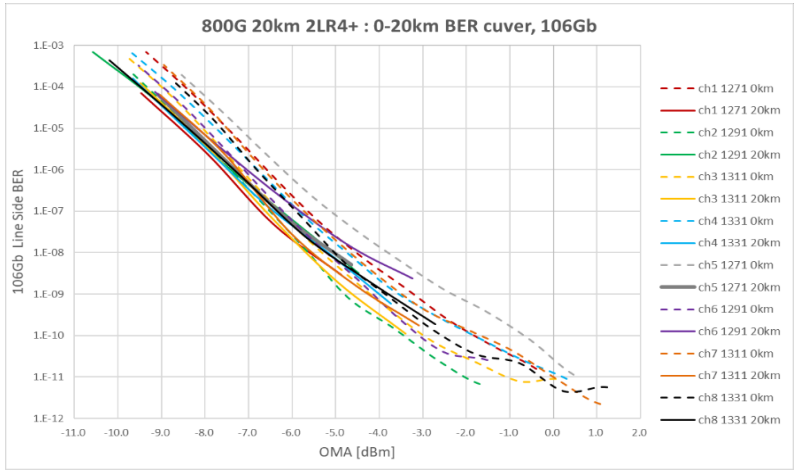
212Gb 10km	1271 without DMD	1271 with DMD	1331 without DMD	1331 with DMD
ER/dB	3.7	3.7	3.7	3.8
TDECQ/dB	NA	0.76	NA	0.83

- **Severe eye diagram distortion at 1271/1331nm due to dispersion, but effectively recovered by DMD.**

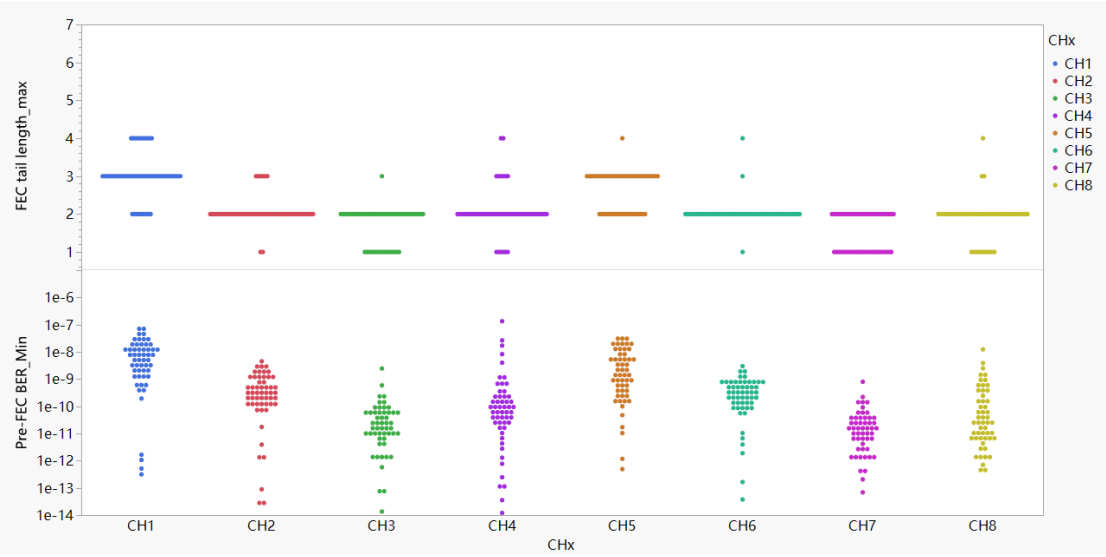
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System Validation:
800G 2LR4+ (20km) Module Performance

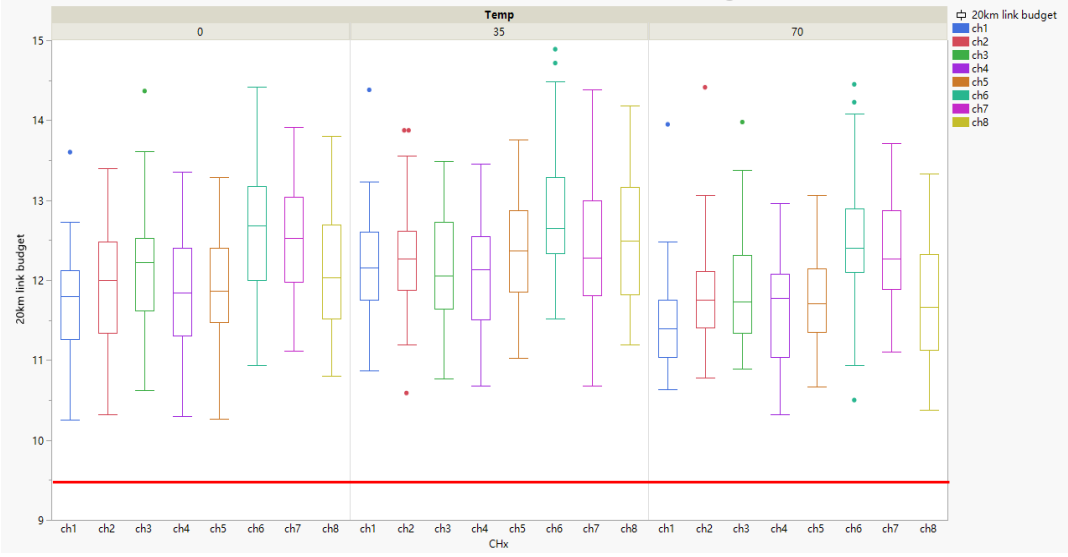
- **BER Curve Overlap:** 20km BER curve perfectly overlaps the 0km curve.
- **Thermal Stability:** Tested across $10°C – 70°C$ thermal cycles; pre-FEC BER remains consistently $< 1e^{-7}$, FEC BIN ≤4.
- **Link Budget Margin:** Robust link budget exceeding **10dB** across all 8 channels.



20km Pre-FEC BER and FEC tail results on TH5 Switch with TC , 10 cycles, 36pcs



20km Link Budget (OMA - 20km Sensitivity@1e-6), 36pcs

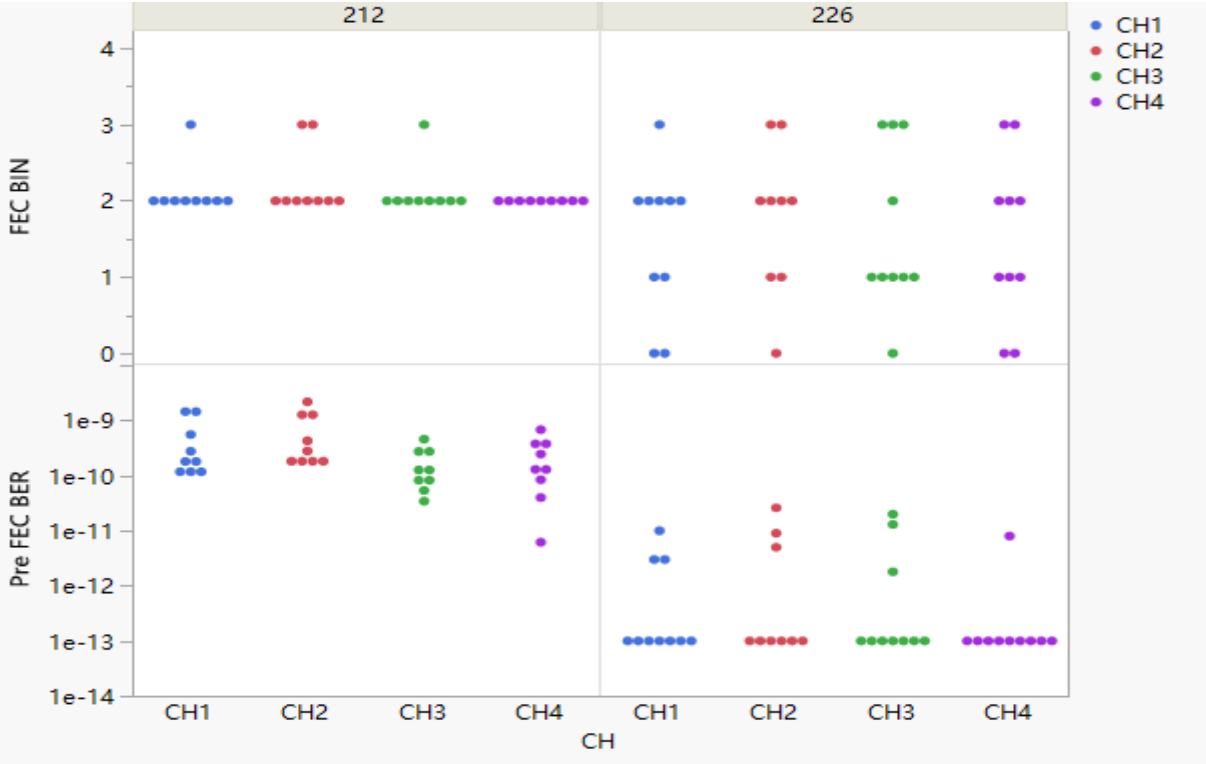


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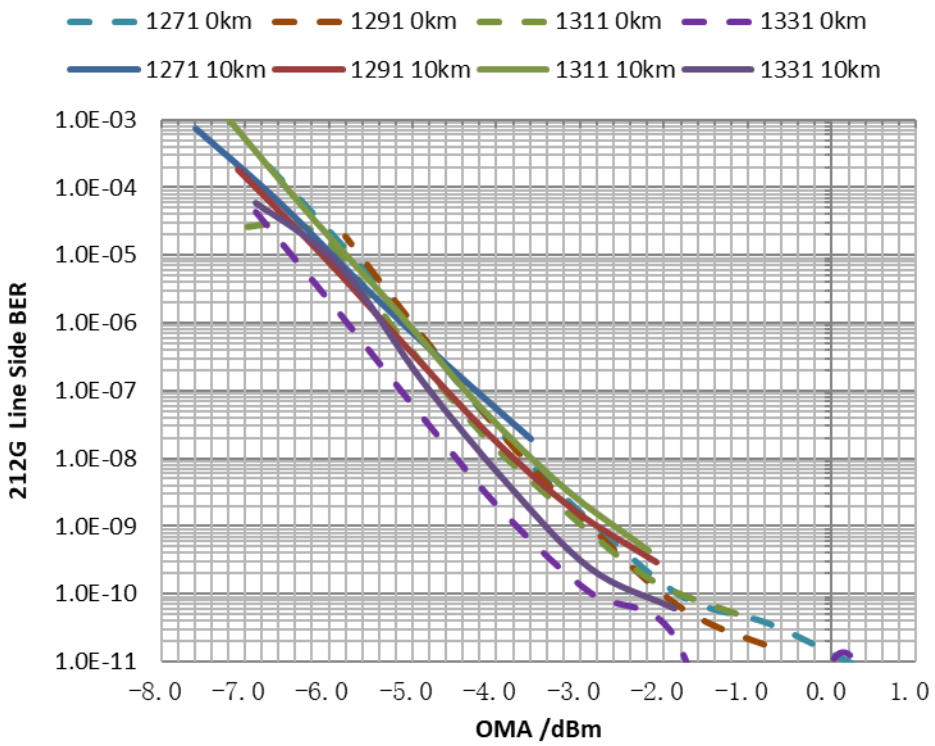
System Validation:
800G FR4 (10km) Module Performance

- **BER Curve Overlap:** 10km BER curve perfectly overlaps the 0km curve.
- **Switch Interop:** Verified over (10 - 70 ℃) on TH5 switch platform, BER and FEC Bin is stable.

10km Pre-FEC BER and FEC tail results on TH5 Switch with TC , 4 cycles, 10pcs



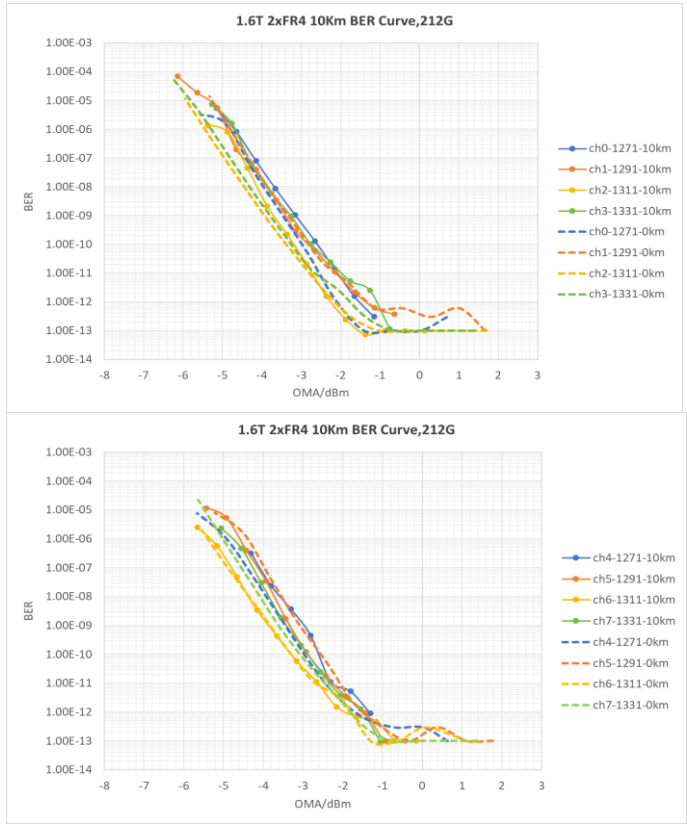
800G FR4 0-10km BER curve,212Gb

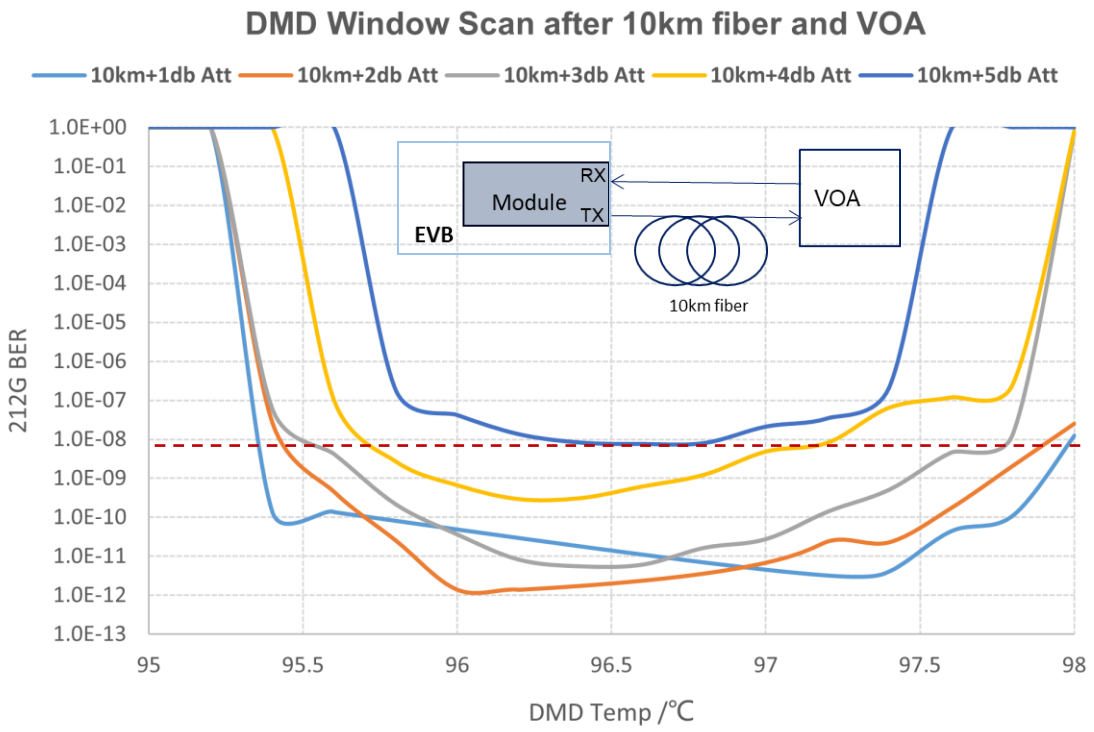


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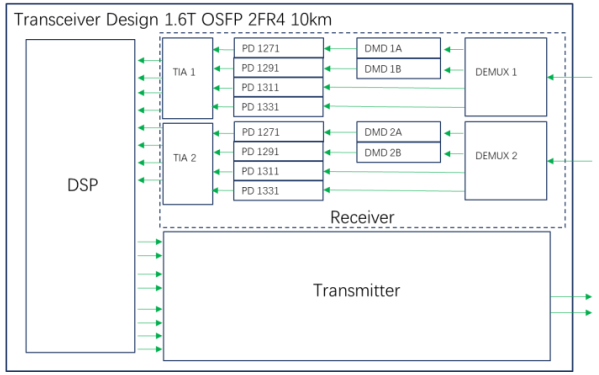
System Validation: 1.6T 2FR4 (10km) Performance

- ✓ **Record Performance:**
  - ✓ $BER < 10^{-11}$@ 212 Gbps over 10km fiber.
  - ✓ $BER < 10^{-12}$@ 226 Gbps with inner FEC enabled.
- ✓ **High Link Budget: 7.2 dB** link budget available.





1.6T 2FR4-10km	212Gb	3KM
0km OMA, dBm	> 4	
OMA-TDECQ	>1.1	>0.1
10km OMA Sensitivity @ 2.4e-6, dBm	<-3.7	<-2
DMD Long-Term Locking Penalty	0.5 dB	0

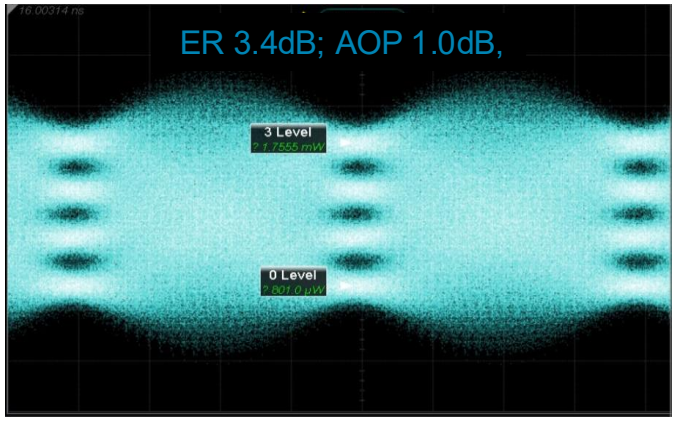


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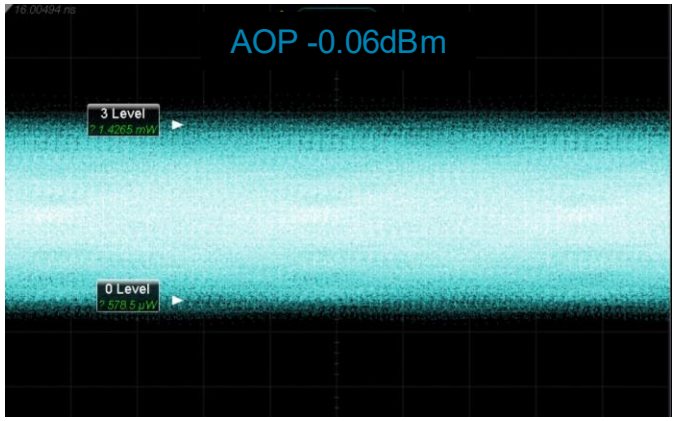
Future-Proofing: 400G/Lane Proof-of-Concept

- **Test Conditions:** Thin-Film Lithium Niobate (TFLN) modulator driven at **190 GBaud PAM4** (380 Gbps rate) at 1271nm.
- **3km Transmission Result**:
  - *Unmanaged 3km:* Signal completely degraded due to severe dispersion.
  - *DMD Compensated:* Clean 4-level PAM4 eye fully restored (ER: 3.3 dB, AOP: -0.6 dBm).
- **Strategic Impact:** Proves CWDM viability for future 3.2T data center links.

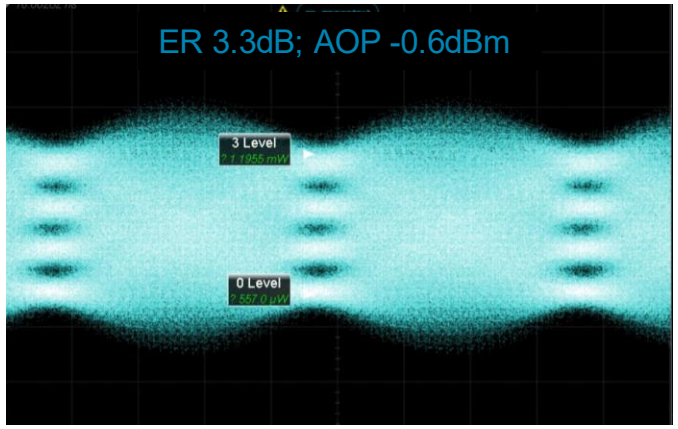
190GBaud, 1271,0km



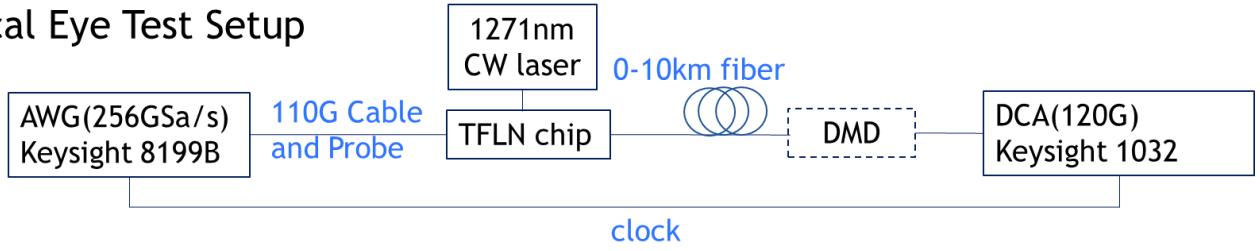
3km

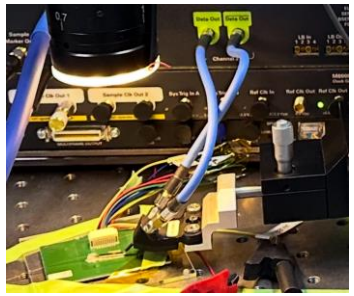


3km + DMD



Optical Eye Test Setup





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Comparative Analysis:
DMD based CWDM, LWDM, Coherent-Lite

DMD demonstrates comprehensive advantages in performance, cost, power efficiency, and compatibility.

	DMD based CWDM	LAN-WDM	Coherent-Lite
Readiness for 200G/L 10km	☺	😐	☺
Potential for 400G/L	☺	😐	☺
Supply Chain (Laser, DSP)	☺	☹	😐
Interoperability with existed FR4	☺	☹	☹
Cost	☺	😐	☹
Power Consumption	☺	☺	😐

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Ecosystem Collaboration & Open Innovation

- **Commitment to Open Standards:** Linktel is opening access to core DMD optical designs to promote industry-wide adoption across OCP SuperPOD fabrics.

- **Protected IP Applications:**
  - U.S. Patent Application No. **19/026,490**
  - U.S. Patent Application No. **19/295,055**
  - U.S. Patent Application No. **19/295,081**

- **Industry Collaboration:** Inviting ecosystem partners, DSP vendors, and cloud operators to co-develop DMD-enabled interconnect solutions.

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Summary & Live Booth Demonstration

Key Takeaways:

- DMD breaks the CWDM dispersion limit for 800G (20km) and 1.6T (10km).
- Zero optical loss, dynamic DSP self-locking, and negligible power overhead.
- Proven 400G/lane scaling path for future 3.2T networks.

Experience the Live Demo:

Location: Linktel Booth **#C84**

Demonstration: Live demo 800G 2LR4+ (20km) and 1.6T 2FR4 (10km)

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